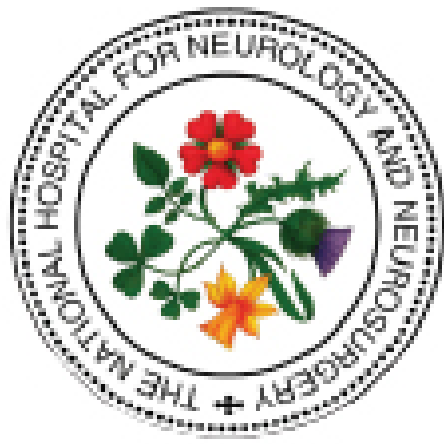




The National Hospital for Neurology and Neurosurgery Neuroscience Critical Care Unit (NCCU) ICP / CPP DIRECTED THERAPY in ACUTE BRAIN INJURY (ABI)

This protocol outlines safe targets based on agreed guidelines, which may be adapted in individual clinical circumstances with the judgement / knowledge of the responsible senior clinician

INTRACRANIAL HYPERTENSION MANAGEMENT

► BASIC NEUROPROTECTIVE MEASURES

Neuroprotective measures involve early, aggressive control of factors implicated in secondary brain injury

A. Artificial Ventilation / Sedation / Analgesia - *propofol & fentanyl infusions*

B. Aim for **PaO₂ 11kPa – 13kPa**, **PaCO₂ 4.5 – 5.0kPa**

► Mandatory correction of hypoxia (defined as SaO₂ <8kPa, cyanosis or apnoea)

► Avoid PaCO₂ <4kPa

► Avoid sustained hyperoxaemia (defined as PaO₂ >16kPa)

C. IV crystalloids aiming for **EUVOLAEMIA** (or slight hypervolaemia)

► Mandatory correction of hypotension (defined as SBP <90mmHg)

► BP management according to disease specific targets but avoiding hypo / hypertension in acute phase

► Ensure adequate fluid resuscitation (guided by FlowTrac monitoring) before starting vasopressors

D. GCS, pupil, limb assessment (clinical assessment depending on level of sedation)

► ICP monitoring recommended in all patients at risk of intracranial hypertension

E. Blood glucose **7 – 12mmol/L**

► Treat hypoglycaemia & hyperglycaemia (see Blood Glucose Protocol)

Head up **30 – 45°**

► Head up 30 - 45° optimal for most ABI patients (if hypotension corrected & spinal injury excluded)

► Head & neck in neutral alignment, avoiding hyperflexion & hyperextension

► Not >45° head up

Temperature – **NORMOTHERMIA** (defined as **37.0 C ± 0.5 C**)

► Treat pyrexia (temperature >37.5C) with antipyretic agent (paracetamol)

► Targeted Temperature Management (TTM) if not resolved 2h after administration (see TTM Protocol)

THERAPEUTIC GOALS ONCE ICP MONITORING COMMENCED

► Target ICP **<22mmHg** (Treat ICP >22mmHg)

The Brain Trauma Foundation Guidelines ICP >22mmHg for prolonged period associated with poorer outcome, therefore BTF guidelines favour ICP <20mmHg & recommend **treating** ICP >22mmHg

► Target CPP **~60mmHg** (Range: 50-70mmHg)

If ICP >22mmHg

STEP 1 - Check

- ✓ Pupil response – changes may prompt urgent CT scan +/- hyperosmolar therapy (see protocol)
- ✓ ETT tapes not impeding venous drainage
- ✓ Head & neck in neutral alignment - return to supine position
- ✓ PaO₂ / PaCO₂ within parameters
- ✓ ICP waveform (? poor trace)

STEP 2 - Ensure

- ✓ Adequate sedation - give bolus & observe effect

STEP 3 - Consider

- Increasing level of sedation - within maximum range
- Adding Midazolam infusion
- NMBA bolus - if effective start infusion
- CT scan

If ICP returns to <22mmHg continue current management, if not

If ICP >22mmHg for >30min

STEP 4 - Consider

- Further MEDICAL MANAGEMENT
 - Increasing sedation to maximum
 - Hyperosmolar therapy
 - EEG to rule out subclinical seizures
- SURGICAL MANAGEMENT
 - Decompressive craniectomy
 - EVD
- ADDITIONAL THERAPEUTIC OPTIONS
 - Therapeutic hypernatraemia: Na⁺ ~155mmol/L

STEP 5

Consider Barbiturate Coma – Thiopentone

STEP 6

Consider review of ICP target

SEDATION and ANALGESIA

Use **SEDLINE** monitor to guide sedative doses

Once burst suppression (state of deep inactivity in the brain / cortical silence) reached, there is no clinical indication to increase sedation further

Infusions above upper limit stated below must be approved by the NCC consultant

PROPOFOL 2% 20mg/mL	0.5mg/kg test bolus 20-75mcg/kg/min infusion For 70kg patient ~ 4-15mL/h
FENTANYL 50mcg/mL	2mcg/kg test dose 2-5mcg/kg/h infusion For 70kg patient ~ 2-7mL/h
MIDAZOLAM 1mg/mL	2mg test dose 30-100mcg/kg/h infusion For 70kg patient ~ 2-7mL/h
Due to potential toxic side effects, duration and dose of any drug must be carefully monitored, Commence at lowest dose and titrate to achieve ICP target	

PROPOFOL

Propofol Infusion Syndrome (PRIS)

Consider in all patients on propofol, especially doses >75mcg/kg/min or any dose when duration exceeds 48h

Screen daily for CK, unexplained acidosis or ECG changes

NEUROMUSCULAR BLOCKING AGENTS

These have no direct effect on ICP but prevent rises produced by coughing on ETT – ensure sedation levels adequate

Atracurium

Give bolus dose / observe effect on ICP
If effective commence infusion (0.5mg/kg/h)

If history of asthma use vecuronium

Propofol and NMBAs not compatible (use separate IV lines)

Seizures difficult to detect if paralysed (may be signaled by bilateral pupillary dilatation, small increase in ABP)

THIOPENTONE

Only used when all other methods to reduce ICP have failed (see Thiopentone protocol)

Treatment of elevated intracranial pressure (ICP) and cerebral perfusion pressure (CPP) is common in patients with neurological injury. ICP / CPP monitoring has been most extensively studied in severe traumatic brain injury (TBI) and is widely used as the standard of care.

It can play a similar role in other patients with Acute Brain Injury (ABI) such as subarachnoid haemorrhage (SAH), intracerebral haemorrhage (ICH) and acute ischaemic stroke (AIS) to help direct medical and surgical therapy.

The threshold that defines intracranial hypertension is uncertain, but it is generally agreed that an **ICP >22mmHg** should be treated and ICP / CPP monitoring are strongly recommended in all patients at risk.

The threshold for ICP in different intracranial pathologies remains controversial. There is no clear evidence for ICP-guided therapy, or whether ICP should be monitored, when the patient has undergone **decompressive craniectomy** (DC) with some studies suggesting that the ICP treatment threshold should be lower in patients with an 'open skull'.

SETTING TARGETS

Monitoring several intracranial variables at the same time (multimodality monitoring – MMM) can complicate decision making in setting targets. Patients with ABI will have prescribed targets based on presenting pathology and their subsequent highest risk – *haemorrhage / ischaemia / oedema etc.* Some patients have conflicting targets and decisions must reflect the key targets to adhere to (**hierarchy of targets**) when achieving all set targets proves difficult. Decisions should always be based on greatest risk.

Consider **phase of injury** when setting targets:

- **Acute phase (~ 24-96h)** set targets (also consider after any subsequent intracranial event)
- **Subacute phase (~ 5-14 days)** consider relaxing targets
- **Post-acute phase (after ~ 14 days)** consider need for targets

There are always risks in **overachieving** as well as in not meeting a set target.

Class of Recommendation

- **Class 1 (Level I) : STRONG**
Recommended (Benefit >>> Risk)
- **Class 2a (Level IIA): MODERATE**
Reasonable (Benefit >> Risk)
- **Class 2b (Level IIB / III): WEAK**
May be reasonable (Benefit ≥ Risk)
- **Class 3: STRONG**
Potential harm (Risk > Benefit)

1. <https://www.braintrauma.org/coma/guidelines>

2. https://www.google.com/url?sa=t&source=web&rct=j&opi=89978449&url=https://bmjopen.bmj.com/content/bmjopen/14/7/e084849.full.pdf&ved=2ahUKFwjv-Osjt6HAXUdZ0EAHdPaASYQFnoECBgQAQ&usg=AOvVaw0Aa18DIz4dZE4zL_ebH3h3

3. https://www.google.com/url?sa=t&source=web&rct=j&opi=89978449&url=https://ccforum.biomedcentral.com/articles/10.1186/s13054-015-0887-8&ved=2ahUKFwjv-Osjt6HAXUdZ0EAHdPaASYQFnoECBgQAQ&usg=AOvVaw1Qkb2W0yB530_LlIArkA7oK